Through the Looking Glass

Mirror Grouping, DNA, and Living Integrity

Arthur Michael Zilkowsky
AMZilkowsky

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An open invitation to see sequences differently

Across the quiet fabric of existence, everything bears a mark.
Not the scars of error, but the signatures of origin—
patterns pressed into matter like light through silk.

What we call anomalies are merely the universe remembering itself,
inverted echoes of perfection,
the negative spaces where truth hides its outline.

Zoom in, and they fragment—
a stutter in code, a tremor in symmetry,
the trembling hand of creation.

Zoom out, and the same distortion resolves into rhythm,
the pulse of a grander design.

These are the watermarks of reality,
etched into DNA, into Pi, into time itself—
proof that even in chaos, the cosmos signs its name.

— AMZilkowsky, Mirage Project, 2025

For every sequence that carries more order than chance would allow,
and for the living coherence that asks to be protected.

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Preface

This book is the result of a long conversation with sequences. It began, as many useful things do, with a simple observation that refused to leave: when identical symbols sit next to each other in a string of digits or letters, something structural is happening. Insert a marker at the exact onset of that repetition and the sequence rearranges itself into non-repeating clusters. Those clusters—mirror groups—turn out to be surprisingly informative.

I first noticed the pattern while exploring the digits of π around the year 2000. The early notes were philosophical and exploratory, written with the excitement of someone who had stumbled on a possible universal grammar. Over the following years the idea matured through application to DNA, through sonification experiments, through visualization systems I called Gene Tablature, and finally through the construction of a modular software framework named Mirage Guardian. Along the way the definition was refined, inconsistencies were resolved, and concrete applications—especially to CRISPR off-target risk and sequence integrity—emerged with measurable force.

What follows is a cleaned, consistent presentation of Mirror Grouping as it stands in late 2025 and 2026. Early speculative language has been tempered where it overreached; formal definitions and working algorithms have been prioritized. At the same time the original sense of wonder has been preserved. Sequences are not merely data. They are patterns that can be seen, heard, measured for coherence, and, in some cases, protected.

I have no large laboratory or institutional funding. The work has been done the efficient way: open sharing, iterative refinement with the help of AI collaborators, public genomic data, and a stubborn belief that simple, interpretable methods still have a place beside deep learning. The CRISPR coherence results, the long non-repeating runs in π, the multi-scale symmetries in mitochondrial DNA, and the living “family” architecture of Mirage Guardian are all products of that approach. The project has been shared on GitHub, in genetics and cybersecurity discussion spaces, and through successive versions of technical papers and color-mapped genomes. Feedback from those exchanges shaped the present form.

This is not a claim that Mirror Grouping solves every problem in genomics or cybersecurity. It is a claim that a simple, deterministic, human-readable transform can reveal structure that other methods sometimes overlook, and that the resulting view of “integrity as lived coherence over time” is worth taking seriously. The book is written for researchers, developers, musicians, and anyone who finds beauty in

the ordered disorder of long symbolic strings.

All core code referenced here is open-source under the MIT license. The project continues to evolve. Readers are invited to experiment, improve, and share. If the mirror shows you something useful, that is the point.

— Arthur Michael Zilkowsky
Dartmouth, Nova Scotia
October 2026

1. Origins — The Discovery in π

The story begins with a number that never ends. π is transcendental; its decimal expansion is infinite and, as far as current mathematics knows, non-periodic. Yet within that expansion, adjacent identical digits occur frequently. Most people treat those runs as noise or as the inevitable consequence of randomness. Mirror Grouping treats them as boundaries.

The First Observations

In the earliest notes, dated around 2000, the idea was expressed more poetically than formally. A mirror group was described as a specifically organized unit, a self-contained form in which no two identical symbols appear consecutively when the sequence is read in order. The mirror itself (the vertical bar |) was the signal that a repeat had been encountered and that a new group must begin. Everything to the right of the mirror dropped, in the original visualization, to the next level—an image of nesting that later became formal Level-2 analysis.

The practical procedure was already present: place a vertical bar whenever any element repeats itself adjacent to the previous one. The result is a sequence of clusters in which no two consecutive symbols are identical. Applied to the first digits after the decimal point of π, the method immediately produces recognizable structure. Longer runs of the same digit force more frequent mirrors and therefore shorter groups. Regions that alternate freely produce longer, more “coherent” groups.

One early fragment of the annotation of π looks roughly like this (the exact grouping depends on where one begins counting, but the principle is constant):

3.14159265358979323846264 3 | 3832795028 | 8419716939 | 937510582097494 | …

Later systematic search, assisted by the same transform, identified unusually long non-repeating stretches. One of these—a 98-digit run free of consecutive identical digits—was independently verified in the first billion digits of π and stands as a minor but concrete number-theoretic observation that emerged directly from the method. The fact that it was found while hunting for maximal coherent intervals, rather than by brute-force enumeration of every possible substring, is itself evidence that the lens is useful.

Early Notes and the Language of Seeds

The 2000 notes are worth hearing in their own voice, not because every claim has been verified, but because they show the original intuition with unusual clarity. A

few representative passages:

> *“What Makes A Mirror Group? 1. Specifically organized unit. A self contained idealistic form. No 2 units are repeated when viewed sequentially. 2. Basic building block exemplified in Pi. 3. A Mirror Group goes from: . to | to () to ( ) propagating infinitely outward, growing in complexity it becomes self-contained, perhaps even self-aware!..lol.”*

> *“If Pi may be referred to as an ‘Infinite’ the level of it’s possible complexity grows in accordance with what is contained within it’s vast realm of the Mirror Groupings. The Mirror Groups make up Pi … Pi itself, as far as current science knows, appears to go on for ever… Voila! Every possible mirror group must be contained in Pi.”*

> *“Every Mirror Group may be coded to specific frequencies. Arranged to suit the ‘senser’. These Group(s) occur in specific locations in Pi. Knowledge of these frequencies and their corresponding locations, is directly related to the knowledge capabilities of the ‘current’ times of inspection.”*

The language of seeds, eggs, pulses, and civilizational awareness is the language of someone who felt that the pattern was deeper than a mere counting trick. Later formalization kept the core mechanism and set aside the stronger metaphysical claims, but the sense that consecutive-repeat boundaries mark something real—something worth listening to—never left the project.

From Philosophy to Practice

From π the method traveled to DNA. The four-letter alphabet of A, T, G, and C is smaller, the biological stakes higher, and the visual and auditory possibilities richer. The same simple rule that carved π into non-repeating clusters began to carve genomes into colored blocks, musical phrases, and measurable coherence scores. The first Gene Tablature color systems appeared, then the first sonification experiments centered near 432 Hz, then the first suspicion that local consecutive-repeat density might correlate with structural risk in guide RNAs.

By late 2025 the method had been formalized, implemented in multiple languages, embedded in a modular guardian architecture, and tested against published CRISPR off-target datasets. The philosophical origin remained visible, but the work had become practical. Technical papers described the algorithm, the mitochondrial case studies, the coherence metric, and the family-consensus security model. Code repositories made the tools reproducible. The rest of this book follows that trajectory: from the looking-glass discovery in π to a set of tools that can be used today.

2. The Method — Formal Definition and Algorithm

Mirror Grouping rests on two complementary measures. Level 1 captures local repetition structure by marking the onset of consecutive identical symbols. Level 2 captures hierarchical symmetry by counting nested matching outer symbols. Together they give a two-level view that is deterministic, efficient, and human-readable.

Level 1: Adjacent-Repeat Segmentation

Let $S = s_1\ s_2\ \ldots\ s_n$ be a sequence over some alphabet Σ. Level-1 Mirror Grouping produces an ordered list of groups $G = (g_1, g_2, \ldots, g_k)$ such that:

- the concatenation of the groups recovers S exactly (the transform is lossless);
- each group g_i is a maximal contiguous substring containing no two consecutive identical symbols;
- a new group begins precisely at the onset of every consecutive repeat.

In other words, a mirror (separator) is inserted between positions $i-1$ and i whenever $s_i = s_{i-1}$. By construction the groups themselves never contain consecutive identical symbols. The boundary is a marker, not a deletion or replacement of any nucleotide or digit. Removing the bars recovers the original sequence without loss or ambiguity.

Simple DNA example. Sequence: A A T G C C
Mirror annotation: A | A T G C | C
Groups: [“A”, “ATGC”, “C”]

Digit example. Input: 2313008908333098611
Annotated: 23130|089083|3|3|309861|1
Groups: [“23130”, “089083”, “3”, “3”, “309861”, “1”]

The two isolated “3” groups are produced by the consecutive pair “33”; each becomes its own maximal non-repeating unit. Symbols outside complete interior groups can be regarded as leading or trailing remainders when that distinction is useful for analysis. In long sequences the distribution of group lengths itself becomes a signature: some regions produce many short groups (high local repetition), others produce fewer, longer groups (higher local coherence).

The algorithm is deterministic, runs in linear time, and uses only constant extra memory beyond the output list. It is trivially invertible. Under a dinucleotide-preserving shuffle the expected boundary density is invariant, giving a simple null model against which observed densities can be compared. That invariance is useful: if a real sequence shows significantly higher or lower boundary

density than its shuffled counterparts, the difference is structural rather than compositional.

Level 2: Recursive Nesting Depth

Once groups (or arbitrary substrings) have been obtained, a second measure captures hierarchical symmetry. For a string $w = w_1\ w_2\ \ldots\ w_m$ define the nesting depth D(w) recursively:

$D(w) = 0$ if $m \leq 1$;
$D(w) = 1 + D(w_2 \ldots w_{m-1})$ if $w_1 = w_m$;
$D(w) = 0$ otherwise.

D(w) simply counts how many layers of matching outer symbols exist before the first mismatch or a single remaining symbol. It is a measure of nested palindromic structure, not of the Level-1 boundary density. The two measures are complementary: Level 1 sees local consecutive repetition; Level 2 sees nested symmetry inside the resulting pieces or inside chosen windows.

Examples:
GC → D = 1 (outer G matches, inner C)
GCCG → D = 2
TATATA → D = 0 (outer symbols differ)
ATATATA → D = 1 (outer A matches; inner TATAT yields another match on T)

In practice, Level-2 analysis is often applied selectively to groups that already look interesting under Level 1, or to sliding windows around known functional sites. Deep nests can be visualized with brackets or progressive color intensity, turning the abstract depth number into something the eye can register at a glance. Future extensions may replace strict symbol identity with DNA-specific complementarity (A–T, G–C), producing “quasi-mirrors” that may be more biologically meaningful.

Properties, Null Models, and Related Methods

Mirror Grouping is deterministic, efficient (O(n) time and space), lossless, and immediately inspectable by eye or ear. It differs from run-length encoding in a fundamental way: RLE collapses runs into counts; MG surfaces the boundaries between runs while preserving every symbol and allowing recursive symmetry analysis inside the resulting pieces. It also differs from classical palindrome finders (Manacher’s algorithm and its relatives) in that the primary segmentation is driven by consecutive-repeat avoidance rather than by exhaustive search for all palindromic substrings. The two can be complementary: MG first carves the sequence into coherent blocks; classical methods can then be applied inside those

blocks or across their boundaries.

The method is rigid by design. Long stretches of a single repeated symbol produce many short groups of length 1. That is not a defect; it is a signal. In DNA such runs are often biologically consequential (homopolymer expansions, slippery sequences, microsatellite instability, etc.). In π they mark the opposite of the long coherent intervals that the method was originally built to find. In both domains the rigidity is the feature that makes the output interpretable and comparable across sequences and organisms.

3. Seeing and Hearing — Visualization and Sonification

Once a sequence has been partitioned into mirror groups, the groups themselves become objects that can be colored, stacked, and sonified. The visual and auditory layers are not decorative; they are load-bearing. They turn an abstract transform into a perceptual instrument that a human can use in real time.

Gene Tablature and the Color System

The Gene Tablature system assigns fixed colors to individual nucleotides and to common dinucleotide “switches.” A typical key, refined over many iterations of genomic mapping, is given in full in Appendix A. In brief: nucleotides receive high-contrast primary colors (T bright yellow, A yellow-green, C cyan/blue, G deep red/pink). Selected dinucleotide pairs that act as binary switches receive secondary colors (AT/TA pink, CG/GC purple, and so on). Palindromes and oscillations are marked in orange or mustard; points of interest in light green; the largest and secondary-largest mirror groups (LMG and sLMG) in deep orange so that the eye is drawn to them immediately.

When a long genomic sequence is rendered under this scheme, the eye perceives regions of high versus low group-length diversity, clusters of nested palindromes, and anomalous stretches that stand out against the background texture. Horizontal layout with vertical stacking of successive groups further improves readability for very long sequences. The same system has been applied to mitochondrial genomes of model organisms (axolotl Ambystoma mexicanum, planarian Schmidtea mediterranea, tardigrade Hypsibius, ancient permafrost nematodes of the genus Panagrolaimus) and to selected nuclear regions, producing distinctive “fingerprints” that are reproducible and comparable. The color maps themselves become part of the scientific record: anyone with the sequence and the open-source code can regenerate them.

Sonification at 432 Hz

Each nucleotide can be mapped to a pure frequency. One working mapping used throughout the Mirage project is a just-intonation set centered near 432 Hz:

$$A \approx 432\ \text{Hz} \quad T \approx 486\ \text{Hz} \quad G \approx 546.75\ \text{Hz} \quad C \approx 512\ \text{Hz}$$

Mirror groups become melodic or rhythmic phrases; the mirrors themselves become rests or phrase boundaries. Nested depth can trigger harmonic overlays (octave, fifth, etc.). The resulting audio is not merely pleasant; it allows pattern recognition by ear. Certain symmetries that are hard to see on a dense page become audible as rhythmic or harmonic regularities. Browser-based live sonifiers

and MIDI exporters have been built so that any user can paste a sequence and hear its mirror structure in real time, with optional looping and harmonic overlays.

The choice of 432 Hz as a reference is partly aesthetic and partly practical: it yields clean integer ratios in the chosen just-intonation set and produces a sonority that many listeners find calming rather than fatiguing during long exploratory sessions. Other tunings are of course possible; the important point is the consistent mapping from symbol to pitch so that structure is preserved in the auditory domain. Early experiments also explored stacking nucleotides on a musical stave “by mass” and using a pentatonic (black-key) framework; those explorations remain available as alternative renderings for users who prefer them.

Why Sensory Feedback Matters

A researcher staring at a million-base FASTA file is limited by the bandwidth of visual search. A researcher who can both see color-coded mirror groups and hear them as musical phrases has two independent channels. Anomalies that are statistically mild but perceptually striking become harder to miss. In the Mirage Guardian architecture this principle is taken further: system integrity itself is made perceptible through a live TUI heartbeat and optional sonification of dissonance. The machine does not merely compute a hash; it can be made to “feel” when its own critical streams lose coherence. That is a different paradigm from purely cryptographic monitoring, and it is one of the distinctive contributions of the project.

The same dual channel supports teaching and public communication. A student who has heard a genome and seen its colored mirror groups has a more durable intuition for local structure than one who has only seen numerical statistics. The aesthetic dimension is not a luxury; it is a way of making the pattern stick in memory and inviting further curiosity.

4. Living Sequences — DNA, Mitochondria, and CRISPR

DNA is a natural home for Mirror Grouping. The alphabet is small, the biological consequences of repetition and symmetry are well studied, and the practical need for lightweight, interpretable features is high. This chapter presents two concrete lines of application: multi-scale symmetry analysis in mitochondrial genomes, and a training-free coherence score that anti-correlates with CRISPR-Cas9 off-target rates.

Mitochondrial Case Studies

Human mitochondrial DNA (mtDNA) is a compact circular genome of approximately 16,569 base pairs. When the first thousand bases are subjected to Level-1 Mirror Grouping followed by recursive Level-2 bracketing, a rich hierarchy of palindromic and quasi-palindromic structures appears. Simple motifs such as [tat], [cac], and [aca] recur; deeper nests of the form <[ca[c]t[c]ac]> also emerge. A substantial fraction of the detected multi-level symmetries overlap the heavy-strand origin of replication (OH) and D-loop regions—areas already known to be structurally dynamic and to interact with proteins such as TFAM.

In one controlled comparison on the initial segment of human mtDNA, Mirror Grouping detected more palindromic events and deeper average nesting than two standard tools (Tandem Repeats Finder and a conventional palindrome analyzer), while showing higher overlap with the OH region. The comparison is not a claim of universal superiority; it is evidence that a symmetry-driven, consecutive-repeat lens surfaces structures that frequency-based or pure-palindrome tools can miss. Limitations remain: the current Level-2 definition uses strict symbol identity rather than complementarity, and functional roles for the observed nests still require experimental validation.

The same pipeline has been applied to other mitochondrial genomes: Ambystoma mexicanum (axolotl), Schmidtea mediterranea (planarian flatworm), Hypsibius (tardigrade), and an ancient Panagrolaimus isolate recovered from Late Pleistocene permafrost roughly forty meters below the surface. Each produces a distinctive visual and statistical texture under Gene Tablature rendering. Some genomes are rich in long alternating stretches; others are dominated by short groups and frequent mirrors. These differences are quantifiable, reproducible, and available for comparative study. The color maps themselves become part of the scientific record.

Coherence Scores and Off-Target Risk

A practical downstream metric is the coherence score derived from Level-1 statistics. Two simple quantities are extracted: the total number of repeat events (separator count) R, and the vector of group lengths L. From these one can form a normalized score that penalizes high local consecutive-repeat density and high variance in group length. Lower coherence corresponds, on average, to higher structural “noise.”

When this score is computed on large published sets of CRISPR guide RNAs with experimentally measured off-target counts (Doench 2016, Kleinstiver 2016, Listgarten 2018, Tsai 2015 clinical-grade libraries, and others—more than twenty-three thousand guides in the aggregated analysis), a clear anti-correlation appears. On the combined benchmarks the Spearman correlation between MG coherence and log(off-target count) is approximately -0.68. For comparison, Shannon entropy alone yields roughly -0.41 and a widely used CFD score roughly -0.59. Deep-learning predictors remain slightly stronger on raw correlation but require large training sets and are less immediately interpretable. Mirror Grouping requires no training data, no reference genome, and no neural network; it is competitive while remaining fully transparent: one can point to the exact mirrors that lowered the score.

A clinical illustration: a spacer consisting of a long run of G’s receives one of the lowest coherence scores in the entire library and is known to be problematic. High-efficiency guides tend to cluster at higher coherence values. The biological intuition is straightforward. Sequences that tolerate high local consecutive repetition are more likely to form secondary structures, seed-repeat interactions, or other self-complementary features that survive Cas9 proofreading and produce off-target cleavage. Mirror Grouping surfaces those features without any sequence alignment.

This does not replace existing off-target predictors. It offers a fast, transparent pre-filter and an independent axis of risk that can be inspected by a human or by a downstream model. Because the method is invertible and zero-dependency, it is also suitable for integration into lightweight clinical or educational pipelines. Further details of the evaluation appear in Appendix D.

What the Patterns Suggest

Taken together, the mitochondrial and CRISPR results suggest that consecutive-repeat structure and nested symmetry are not merely mathematical curiosities. They appear to track biologically relevant properties: the location of replication origins, the risk of off-target editing, and the overall “texture” of a genome under a simple, reproducible lens. Whether these patterns ultimately

reflect selection for structural stability, constraints of replication or transcription, or deeper principles of sequence organization is a question for further experimental work. Mirror Grouping supplies a practical way to pose the question and to generate testable predictions.

5. The Guardian — Architecture of Living Integrity

Mirror Grouping began as a sequence transform. It grew into a modular system. Mirage Guardian treats analysis, visualization, sonification, cryptographic integrity, and monitoring as members of a single “family” that continuously cross-validates itself. The guiding principle is that integrity should be lived and felt, not only computed.

Design Principles

Five principles shape the architecture:

1. Self-sufficiency — every module can operate and test itself in isolation.
2. Mutual awareness — modules broadcast state over a lightweight context bus.
3. Consensus truth — system health is declared only when the family agrees.
4. Transparency — operations are logged and cryptographically verifiable.
5. Perceptibility — integrity is not only hashed; it can be seen and heard (TUI heartbeat, sonification of dissonance).

The slogan that captures the intent is “integrity as lived coherence over time.” A system that merely computes a hash can be correct yet feel dead. A system that continuously measures the mirror structure of its own critical streams and surfaces deviations as sensory events is closer to a living guardian.

Core Components and Consensus

The principal components are:

• Core Engine — the deterministic Mirror Grouping transform and recursive symmetry mapper.
• Guardian Context — a minimal pub/sub bus (< 2 KB of core logic) for inter-module state sharing.
• Modules — Visualizer, Sonifier, CRISPR Analyzer, Crypto Store, Watchdog, and others as needed.
• Consensus Heartbeat — aggregates module health and agreement status, displayed live in a textual UI.
• Security Fabric — SHA-256 hashing, signed family logs, Fernet-based encrypted secret storage.

Because the core transform is O(n) and zero-dependency, the entire family can run on modest hardware and still provide real-time feedback. Applications demonstrated so far include ransomware-style tampering detection (sudden collapse of coherence in monitored files or streams), self-verification of analysis pipelines, and pre-screening of guide-RNA libraries. The consensus protocol is intentionally lightweight: modules share signed status; disagreement triggers an

immediate watchdog alert that is both logged and made perceptible. Recovery can draw on the crypto store to restore known-good state.

Integrity as Lived Coherence

The deeper claim is architectural as well as technical. Most security systems treat integrity as a static property verified by cryptographic means. Mirage Guardian treats it as a dynamic, observable process. When the mirror structure of a critical sequence changes in ways that the family has not agreed upon, the system experiences dissonance. That dissonance is not hidden inside a log file; it is surfaced so that a human operator (or another module) can respond. The same principle that makes a genome's texture audible and visible is applied to the system's own health.

The architecture is intentionally open and modular. New modules can join the family provided they respect the consensus and logging conventions. The philosophical claim—that coherence can be felt as well as computed—remains the north star. Whether this model scales to large distributed systems is an open question; the local, human-scale version already works and already changes how integrity is experienced. The closing line that has accompanied the project is worth repeating here: "Mirage doesn't fight attacks. It listens for when the universe forgets its own name."

6. How to Begin — A Practical Path

The best way to understand Mirror Grouping is to run it. This short chapter offers a minimal path from zero to first insight.

Step 1 — Obtain a sequence

Any symbolic string will do. For a first DNA experiment, download a short mitochondrial sequence from NCBI (for example the first 500–2000 bases of human mtDNA NC_012920.1, or any of the model-organism mtDNAs mentioned earlier). For a numerical experiment, take the first few hundred digits of π after the decimal point. For a musical experiment, convert a short melody to a sequence of note names or MIDI numbers.

Step 2 — Apply Level 1

Use the core function given in Appendix C (or any of the open-source implementations). The output is a list of groups. Print them with "|" separators so the eye can see the boundaries. Count the groups, compute their lengths, and note where the shortest and longest groups fall.

Step 3 — Add color or sound (optional but recommended)

Map the nucleotides (or digits, or notes) to the Gene Tablature colors or to the 432 Hz frequency set. Even a simple spreadsheet coloring or a few lines of browser JavaScript will make the texture visible or audible. Listen for rhythmic regularities; look for clusters of short groups or for long coherent blocks that stand out.

Step 4 — Compare and question

Shuffle the sequence while preserving dinucleotide frequencies and re-run the transform. Does the real sequence produce a different group-length distribution? Apply Level 2 to a few of the longer groups. Do nested symmetries appear near known functional sites? If you are working with guide RNAs, compute the simple coherence score and see how it ranks known high- and low-risk spacers.

That is enough to begin. The rest of the book—and the open-source repositories—exist to support whatever questions arise once the mirror is in hand.

7. Reflections — Philosophy, Open Questions, and Invitation

Mirror Grouping does not claim to be a universal decoder of mysteries. It claims only that a simple, deterministic way of marking the onset of consecutive repetition yields a useful, human-scale view of structure in long symbolic sequences. That view has already proven practical in DNA analysis, CRISPR safety scoring, visualization, and sonification. It has also proven fertile for systems thinking about integrity.

The early poetic language of 2000—self-contained units, frequency codes, civilizational awareness of π—need not be discarded. It can be read as an intuition that patterns of repetition and symmetry are fundamental, and that a civilization's depth of engagement with those patterns is one measure of its maturity. Whether one prefers the language of mathematics, biology, music, or metaphysics, the same objects are under discussion: the places where a sequence refuses to repeat itself consecutively, and the nested symmetries that appear inside those places.

Several open questions remain active research territory:

- How should Level-2 nesting be extended to DNA-specific complementarity (A–T, G–C) rather than strict symbol identity? Quasi-mirrors of that kind may be more biologically meaningful than perfect palindromes of identical bases.
- Can recursive mirror depth serve as a robust, hard-to-game entropy estimator for streams that must resist adversarial manipulation?
- What is the precise statistical relationship between mirror-group statistics and higher-order chromatin structure or RNA secondary structure in vivo?
- How far can the "family consensus" model scale in distributed or adversarial settings without losing its human-perceptible character?
- Are there optimal alphabets or symbol equivalences for particular domains—protein sequences, musical MIDI event streams, network packet headers, or binary firmware images?
- Do the long coherent intervals in π (and in other constants) have any deeper number-theoretic significance, or are they simply the expected extremes of a random-like process viewed through this particular lens?

The invitation is simple. Take a sequence you care about—genomic, numerical, musical, or textual—run the transform, look at the groups, listen to them, measure their coherence, and ask what the resulting texture tells you. The method is free, the code is open, and the mirror is patient. If the patterns you find prove useful for safer editing, clearer visualization, more honest integrity monitoring, or simply a deeper appreciation of the order hidden in long strings, then the work has done what it set out to do.

May the sequences you study reveal more order than chance would predict, and may the tools we build help protect the living coherence that makes such order possible.

Glossary

Coherence score
A scalar derived from Level-1 statistics (repeat density and group-length variance) that quantifies how “clean” or “noisy” a sequence appears under Mirror Grouping. Higher values indicate lower consecutive-repeat density and more balanced group lengths. Used as a training-free feature for CRISPR off-target risk assessment.

Gene Tablature
A color-coding and layout system for rendering Mirror-Grouped DNA sequences so that nucleotides, dinucleotide switches, palindromes, points of interest, and largest mirror groups are visually distinct. Designed for both screen and print readability of long sequences.

Largest Mirror Group (LMG)
The longest group produced by Level-1 segmentation of a given sequence. Secondary LMGs (sLMGs) are the next-longest distinctive groups; both are often highlighted because they stand out against the surrounding texture.

Level 1 (MG Segmentation)
The primary transform that inserts a separator at the onset of every consecutive identical symbol, producing maximal groups that contain no consecutive repeats. Lossless, deterministic, O(n).

Level 2 (Nesting Depth)
A recursive measure that counts matching outer symbols in a substring, capturing nested palindromic or quasi-palindromic structure. Complementary to Level 1.

Mirror / Separator
The boundary marker (commonly written “|”) placed between positions i−1 and i when $s_i = s_{i-1}$. It is a pure annotation; no symbol is deleted or altered.

Mirror Group
A maximal contiguous substring containing no two consecutive identical symbols; the basic unit produced by Level-1 segmentation.

Mirage Guardian
A modular software framework built around Mirror Grouping that adds visualization, sonification, cryptographic integrity checking, and family-style consensus monitoring. Designed so that integrity can be perceived as well as computed.

Sonification
The mapping of sequence elements or mirror groups to musical frequencies (commonly a 432 Hz-centered just-intonation set) so that structure can be perceived by ear. Nested depth may trigger harmonic overlays.

Family consensus
The Mirage Guardian protocol in which modules share signed status and reach agreement on system health. Disagreement produces immediate, perceptible alerts rather than silent failure.

Remainder
A leading or trailing fragment that does not form a complete interior mirror group under the segmentation rule; sometimes useful to distinguish from full groups when analyzing boundary effects.

Appendix A: Color Key for Gene Tablature

The following key has been used across multiple genomic visualizations. Exact RGB or ANSI values may be adjusted for display media or accessibility needs, but the relative assignments remain stable so that maps produced at different times remain comparable.

Nucleotides
T — Bright Yellow
A — Bright Green / Yellow-Green
C — Bright Cyan / Blue
G — Deep Red / Pink

Selected dinucleotide switches
AT / TA — Pink
CG / GC — Purple
TG / GT — Yellow-Green
CT / TC — Sky Blue
GA / AG — Blue
CA / AC — Yellow

Structural markers
Palindromes and oscillations — Orange / Mustard Yellow
Points of interest — Light Green
LMGs and sLMGs (largest and secondary largest Mirror Groups) — Deep Orange

When rendering, consecutive groups are often stacked vertically or placed in a horizontal flow with clear separators so that the eye can track group boundaries at a glance. Nested Level-2 structures may be indicated by brackets, by progressive intensity of the special-marker color, or by both. The system is intentionally simple enough that a researcher can implement a readable version in a few dozen lines of code, yet rich enough that large genomes become navigable by texture rather than by raw coordinate alone.

Appendix B: Worked Examples and Sequences

B.1 Short DNA fragment

Input (illustrative): CTCTTTTAGTTTATATAAAATATACTTTGAAGCAGTTAGAAT

One possible Level-1 presentation (groups separated by |):
CTCT | T | T | TAGT | T | TATATA | A | A | ATATA | ACT | T | TGA | AGCG | GT | TAGA | AT

Notice how runs of T and of A force short groups, while mixed regions produce longer coherent blocks. Under Gene Tablature coloring the short groups of identical bases stand out immediately, and any nested Level-2 structures inside the longer groups can be bracketed for further inspection.

B.2 Consecutive-repeat illustration with digits

Input: 2313008908333098611
Annotated: 23130|089083|3|3|309861|1
Groups: [“23130”, “089083”, “3”, “3”, “309861”, “1”]

The consecutive pair “33” produces two successive length-1 groups. This is the canonical behavior of the transform: every onset of consecutive identity becomes a boundary.

B.3 π and the search for long coherent intervals

The early exploration of π focused on locating long stretches free of consecutive identical digits. One such stretch, later verified independently in the first billion digits, contains 98 consecutive digits with no two identical neighbors. The discovery was guided by the same maximal-group search that the Level-1 algorithm formalizes. Whether longer runs exist deeper in the expansion is an open computational question; the method supplies an efficient way to look.

B.4 Mitochondrial texture (qualitative)

When the first kilobase of human mtDNA is rendered under Gene Tablature after Mirror Grouping, the eye sees a mixture of short groups (often associated with local homopolymers or simple repeats) and longer alternating stretches. Nested brackets cluster preferentially near the OH and D-loop regions in the analyses performed to date. Comparative maps of planarian, tardigrade, and ancient nematode mtDNA show organism-specific balances between these two regimes. The maps are reproducible from the public sequences and the open-source transform; they

constitute a new kind of comparative character.

B.5 Axolotl mitochondrial opening (illustrative fragment)

From Ambystoma mexicanum mitochondrion (NC_005797.1), an early fragment after Mirror Grouping and color annotation begins with patterns such as:

T | TA | ATGTAGCT | TA | A | ATA | A | AGTGTGACACTGA | …

Even this short stretch already shows the alternation of short groups and longer mixed blocks that characterizes many mitochondrial openings under the transform. Full maps are available in the project materials for those who wish to examine the complete texture.

Appendix C: Core Pseudocode

The following Python-style pseudocode implements Level-1 Mirror Grouping. It is deterministic, O(n), and produces the list of groups whose concatenation recovers the original sequence.

```
def group_sequence(seq: str) -> list[str]:
    """Level-1 Mirror Grouping: maximal groups with no consecutive repeats."""
    if not seq:
        return []
    seq = seq.strip()
    groups = []
    current = seq[0]
    for ch in seq[1:]:
        if ch == current[-1]:          # onset of consecutive repeat
            groups.append(current)
            current = ch
        else:
            current += ch
    groups.append(current)
    return groups

# Example
# group_sequence("AATGCC")  →  ["A", "ATGC", "C"]
```

Recursive nesting depth (Level 2) can be implemented separately on any substring or group:

```
def nesting_depth(w: str) -> int:
    """Count nested matching outer symbols."""
    if len(w) <= 1:
        return 0
    if w[0] == w[-1]:
        return 1 + nesting_depth(w[1:-1])
    return 0
```

A minimal coherence-score sketch (illustrative):

```
def coherence_score(seq: str) -> float:
    groups = group_sequence(seq)
    if not groups:
        return 1.0
    lengths = [len(g) for g in groups]
    R = len(groups) - 1          # approximate separator count
    mean_len = sum(lengths) / len(lengths)
    # simple penalty for high repeat density and high length variance
    # (production versions use calibrated normalization)
    ...
```

Production implementations add color annotation, full statistics collection, optional DNA-specific complementarity checks for Level 2, and streaming variants for very long sequences. The core logic remains the functions above. Reference implementations in Python, JavaScript, and other languages are available in the

open-source Mirage repository.

Appendix D: Notes on the CRISPR Evaluation

The coherence-score results summarized in Chapter 4 were obtained by applying Level-1 Mirror Grouping to published guide-RNA sequences that already carried experimentally measured off-target counts. The principal public sources include the Doench 2016, Kleinstiver 2016, Listgarten 2018, and Tsai 2015 clinical-grade libraries, among others, totaling more than 23,000 guides in the aggregated analysis.

For each guide the transform yields a separator count R and a vector of group lengths. A simple normalized score is formed that decreases with higher R and with higher relative variance of the lengths. The resulting scores anti-correlate with log-transformed off-target counts (Spearman $r \approx -0.68$ on the aggregate set). The correlation is visible on individual datasets as well and outperforms both raw Shannon entropy and a standard CFD score on the same collections, while remaining training-free and fully interpretable.

These numbers should be read as evidence of utility, not as a claim that Mirror Grouping replaces state-of-the-art deep-learning predictors. The practical value lies in speed, transparency, and the ability to point to the exact consecutive repeats that drove a low score. Integration as a pre-filter or as an additional feature in larger models is a natural next step that the community is invited to explore. Full reproduction materials (scripts, intermediate tables, and exact guide lists used) are intended for release alongside the open-source code so that independent verification is straightforward.

Appendix E: Selected Early Notes (2000)

The following passages are taken from the original 2000 notes with only light modernization of punctuation for readability. They are included so that the philosophical root of the work remains visible alongside the later formalization.

> *“What Makes A Mirror Group?*
>
> *1. Specifically organized unit. A self contained idealistic form. No 2 units are repeated when viewed sequentially.*
>
> *2. Basic building block exemplified in Pi.*
>
> *3. A Mirror Group goes from: . to | to () to ( ) propagating infinitely outward, growing in complexity it becomes self-contained, perhaps even self-aware!..lol.*
>
> *4. If Pi may be referred to as an “Infinite” the level of it’s possible complexity grows in accordance with what is contained within it’s vast realm of the Mirror Groupings. The Mirror Groups make up Pi … Pi itself, as far as current science knows, appears to go on for ever… Voila! Every possible mirror group must be contained in Pi.*
>
> *5. Every Mirror Group may be coded to specific frequencies. Arranged to suit the “senser”. These Group(s) occur in specific locations in Pi. Knowledge of these frequencies and their corresponding locations, is directly related to the knowledge capabilities of the “current” times of inspection.*
>
> *6. A mirror | occurs in Pi or other applicable numerical sequences, whenever any element repeats itself.”*

These notes are the seed, not the finished tree. The formal definitions, algorithms, biological applications, and system architecture developed between 2025 and 2026 are the growth that followed. Keeping both in view is part of the integrity of the project.

Acknowledgments & Further Resources

This work grew through sustained conversation with sequences and with collaborating AIs (notably iterations of ChatGPT/Luna, Grok, and Gemini) that helped refine definitions, test edge cases, draft technical papers, and keep the project moving under resource constraints. The scientific and artistic responsibility remains with the author.

Open-source code, additional examples, live sonification demos, and evolving documentation are available under the MIT license in the Mirage project repository associated with the author. Readers are encouraged to experiment, fork, improve the tools, and share both successes and failures.

Special thanks are due to every researcher who has shared genomic data publicly, to the maintainers of π digit archives that made independent verification of long non-repeating runs possible, and to the broader community that still believes simple, interpretable methods have a place beside deep learning. Thanks also to the organisms whose mitochondrial genomes provided such rich material for comparison—axolotl, planarian, tardigrade, and the ancient nematode recovered from permafrost—each of which left a distinctive watermark under the mirror.

“Mirage doesn’t fight attacks. It listens for when the universe forgets its own name.”

End of book
Through the Looking Glass — Mirror Grouping, DNA, and Living Integrity
Arthur Michael Zilkowsky · Mirage Project · 2026

All core methods and code open under the MIT license. Share freely.